UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/573,569	12/22/2023	Jinyao TANG	04435/010032-US1	7352

76808 7590 05/18/2026

Leason Ellis LLP
One North Lexington Avenue, Suite 1200
White Plains, NY 10601-1526

EXAMINER
GROOMS, NOA WILLIAM FRAN

ART UNIT	PAPER NUMBER
1759	

NOTIFICATION DATE	DELIVERY MODE
05/18/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspto@leasonellis.com

Office Action Summary	Application No. 18/573,569	Applicant(s) TANG et al.	
	Examiner Noa W Grooms	Art Unit 1759	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 22 December 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-8 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 22 December 2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 26 Mar 2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Acknowledgment is made of applicant's claim for foreign priority under 35 U.S.C. 119 (a)-(d). The certified copy has been filed in parent Application No. PCT/CN2022/097814, filed on June 9, 2022.

Drawings

3. The drawings are objected to as failing to comply with 37 CFR 1.84(p)(5) because they include the following reference character(s) not mentioned in the description: **Fig. 4A member 42 is undefined in specification**. Corrected drawing sheets in compliance with 37 CFR 1.121(d), or amendment to the specification to add the reference character(s) in the description in compliance with 37 CFR 1.121(b) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

4. The drawings are objected to because **Figs. 2C-2H a legend should be provided to indicate 1 vs 5 vs 10 vs 20 mW/cm2**. Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

5. The disclosure is objected to because of the following informalities: Page 3 line 10 "color-shitting paint" should be corrected to "color-shifting paint".

Appropriate correction is required.

Claim Rejections - 35 USC § 112

6. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

7. **Claim 6 is** rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

8. **Claim 6** recites the limitation "other redox couples" in ending of claim sentence. There is insufficient antecedent basis for this limitation in the claim. The claim is not clear is solvent is considered a redox couple. The claim is also not clear if multiple solvents and multiple redox couples are required. The claim is interpreted to be directed to further comprising a solvent or other redox couple based on the specification disclosure of "solvents and other redox couples can be used to improve the resolution."

9. The following is a quotation of 35 U.S.C. 112(d):

> (d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

> Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

10. **Claim 5 is** rejected under 35 U.S.C. 112(d) or pre-AIA 35 U.S.C. 112, 4th paragraph, as being of improper dependent form for failing to further limit the subject matter of the claim upon which it depends, or for failing to include all the limitations of the claim upon which it depends. Claim 5 is dependent on claim 4 which inherits the dependencies and limitations of claims 1-3.

In claim 3, the "at least three colored dyes" are defined as "organic materials". Claim 4 defines the dyes as: SQ2, LEG4, and L0, which are all properly organic materials. However, claim 5 states that "at least one dye is made of inorganic material". Based on the language of claim 4 which dictates the dyes as being solely SQ2, LEG4, and L0, there is no basis of inorganic material being included in any of the dyes. Therefore, claim 5 does not further limit the claims on which it depends. Applicant may cancel the claim(s), amend the claim(s) to place the claim(s) in proper dependent form, rewrite the claim(s) in independent form, or present a sufficient showing that the dependent claim(s) complies with the statutory requirements.

Claim Rejections - 35 USC § 103

11. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

12. **Claims 1-3 and 7** are rejected under 35 U.S.C. 103 as being unpatentable over Zhang et al (CN112208237A) in view of Jing Zheng, et al. "Orthogonal navigation of multiple visible-light-driven artificial microswimmers," Nature Communications Vol. 8, No. 1, November 10, 2017, pages 1-7.

a. Regarding **claim 1**, Zhang discloses a **photochromic ink (liquid)** composed of **three dyes** corresponding to **cyan, fuschine (also referred to as "red purple dye" which "is called magenta" under the "preparation" section), and yellow** whereby **using different absorption spectrum of each dye to control each color channel in a photochromic coating**. By modifying the light source (see explanation under material 2

in document, preparation of photochromic coating, and Fig. 4), the **full gamut of color output can be obtained**. However, Zhang does not disclose use of a colloid imbued with the dyes. Zheng et al **teach imbuing TiO_2 nanowires with individual organic (RGB) dyes** in a **liquid solution (thus a colloid, Fig. 1)** which are also capable of achieving the **full color gamut (Fig. 2**). These imbued nanowires can serve as "microswimmers", essentially navigating in 3D throughout a solution in response to light specific to the imbued dye (Fig. 4). The sensitized microswimmers can be independently propelled and navigated (see Discussion). It would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to substitute the colloidal-based dye in the composition taught by Zhang to produce a photochromic coating with more dynamic translational light-specific responses. Therefore, Zhang and Zheng teach the claimed "**A chromic liquid capable of responding to and reproducing external optical color illumination comprising a colloid imbued with at least three colored dyes selected on the basis of a color system in order to provide a full gamut of color output.**".

b. Regarding **claim 2**, Zhang and Zheng teach the chromic liquid of **claim 1**. Furthermore, Zhang discloses **dyes corresponding to cyan, magenta, and yellow.** When all **three color channels are completely saturated**, the **obtained color is black and thus CMYK color system** (see "Basic Working Principles"). Therefore, Zhang and Zheng teach the claimed "**The chromic liquid of claim 1 wherein the dyes are selected based on the CMYK color system**.".

c. Regarding **claim 3**, Zhang and Zheng teach the **chromic liquid of claim 2**. Zhang does not disclose the identity of the dyes outside of the color spectra. Zheng teaches that the titania is loaded with a "**photosensitizer (organic dyes)"** in the final paragraph of the

introduction. Under the first paragraph, Zheng discloses use of **N719, D5, and SQ2 which are all organic dyes**. It would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to use any combination of CMY organic dyes with a predictable result of creating a chromic liquid. Thus, Zhang and Zhen satisfy the claimed "**The chromic liquid of claim 2 wherein the dyes are organic materials**.".

d. Regarding **claim 7**, as described in the rejection of **claim 1**, Zhang discloses a **photochromic ink (liquid)** composed of **three dyes** corresponding to **cyan, fuschine (also referred to as "red purple dye" which "is called magenta" under the "preparation" section), and yellow** whereby **using different absorption spectrum of each dye to control each color channel in a photochromic coating**. By modifying the light source (see explanation under material 2 in document, preparation of photochromic coating, and Fig. 4), the **full gamut of color output can be obtained**. Furthermore, Zhang discloses use **of a projector which transmits the "texture generated by the computer to the canvas" whereby the canvas is the photochromic liquid/ink**. The **user can "use handheld UV projector on the photo-chromic canvas 'picture'", thus a system for projecting a color image onto the photochromic material**. Under "modomodify projector" section, Zhang describes use of an **existing DLP projector because it already comprises three LED light sources of different colors (RGB)** and **use of an optical filter for each LED between collimators (see Fig. 4, 'transparent' containers depicted for Fig. 5**). Additionally, **Fig. 3 shows emission spectra of the projector which are "spiked" due to the color optical filter**. However, Zhang does not disclose use of a colloid imbued with the dyes. Zheng et al **teach imbuing TiO_2 nanowires with individual organic (RGB) dyes** in a **liquid solution (thus a colloid,**

Fig. 1) contained in transparent containers which are also capable of achieving the **full color gamut (Fig. 2)**. These imbued nanowires can serve as "microswimmers", essentially navigating in 3D throughout a solution in response to light specific to the imbued dye (Fig. 4). The sensitized microswimmers can be independently propelled and navigated (see Discussion). It would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to substitute the colloidal-based dye in the composition taught by Zhang to produce a photochromic coating with more dynamic translational light-specific responses. Thus, Zhang and Zheng satisfy the claimed "**A color display comprising:a transparent container containing a colloid imbued with at least three colored dyes selected on the basis of a color system;a first projector for projecting a color image onto the container for a fixed period of time; and three light sources of different color which direct their light through color related spike optical filters and then into the container for a certain period of time**.".

13. **Claims 4-6 and 8** are rejected under 35 U.S.C. 103 as being unpatentable over Zhang in view of Zheng as applied **to claims 3 and 7** above, and further in view of Takayuki Kitamura, et al. "Phenyl-Conjugated Oligoene Sensitizers for TiO2 Solar Cells," Chemistry of Materials 2004 16 (9), 1806-1812 DOI: 10.1021/cm0349708 and Kenji Kakiage, et al. "Highly-efficient dye-sensitized solar cells with collaborative sensitization by silyl-anchor and carboxy-anchor dyes," Chem. Commun., 2015,51, 15894-15897 https://doi.org/10.1039/C5CC06759F.

a. Regarding **claim 4**, Zhang and Zheng teach the **chromic liquid of claim 3.** Zheng **teaches use of SQ2 and that any organic dyes or photosensitizer can be used with TiO_2 in a colloid** but neither disclose use of LEG4 or L0 dyes. Kitamura characterizes phenyl-conjugated oligoene dyes adsorbed on TiO_2 films. In scheme 1, the **structure of**

1b is that of **4-diphenylamino phenylcyanoacrylic acid (L0)**. Kitamura discloses in Table 1, the **emission of dye 1b is around 550nm or yellow-green**. Kakiage discloses use of **LEG4 adsorbed onto a TiO_2 electrode** to improve electron injection and conversion efficiency. Thus, it would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to substitute the LEG4 and 4-diphenylamino phenylcyanoacrylic acid (L0) dyes for cyan and yellow, respectively, along with use of SQ2 as taught by Zheng into the composition of Zhang to produce a photochromic liquid. Therefore, Zhang, Zheng, Kitamura, and Kakiage teach the claimed "**The chromic liquid of claim 3 wherein the colloid is a TiO_2 colloid and the dyes are:(a) SQ2 (5-carboxy-2-[[3 -[(2,3-dihydro- 1,1 -dimethyl-3-ethyl- 1H-benzo[e]indol-2- ylidene)methyl]-2-hydroxy-4-oxo-2-cyclobuten- 1-ylidene]methyl] -3 ,3-dimethyl- 1-octyl-3H-indolium), (b) LEG4 (3- {6-{4-[bis(2',4'-dibutyloxybiphenyl-4-yl)amino-]phenyl}-4,4-dihexyl- cyclopenta-[2, 1-b: 3,4-b']dithiophene-2-yl}-2-cyanoacrylic acid) and (c) L0 (4-(diphenylamino) phenylcyanoacrylic acid); and wherein the liquid reproduces cyan, magenta and yellow color or a mixture thereof in response to exposure to illumination by light of various colors.**"

b. Regarding **claim 5**, Zhang, Zheng, Kitamura, and Kakiage **teach the chromic liquid of claim 4**. Furthermore, Kakiage teaches use of a **silyl-anchor dye ADEKA-1 (made of inorganic material) bonded in coordination with LEG4**. Thus, Zhang, Zheng, Kitamura, and Kakiage teach the claimed "**The chromic liquid of claim 4 wherein at least one dye is made of inorganic material**."

c. Regarding **claim 6**, Zhang, Zheng, Kitamura, and Kakiage **teach the chromic liquid of claim 4.** Further, both Zheng and Kitamura describe that these adsorbed dyes

can be typically implemented into dye-sensitized solar cells. Zheng and Kitamura both disclose **that these set ups include a redox couple (I^-/I_3^-) in the electrolyte**. Additionally, Zhang describes that **the photochromic dye is "directly dissolved in the transparent paint" and thus contains a solvent**. Thus, Zhang, Zheng, Kitamura, and Kakiage satisfy the claimed "**The chromic liquid of claim 4 further comprising solvents and other redox couples.**".

d. Regarding **claim 8**, Zhang and Zheng teach the display of **claim 7**. As described in the rejection of **claim 4**, Zheng also **discloses use of a TiO_2 colloid and where one dye is SQ2**. Zheng states that any organic dye may be used. Kitamura characterizes phenyl-conjugated oligoene dyes adsorbed on TiO_2 films. In scheme 1, the **structure of 1b** is that of **4-diphenylamino phenylcyanoacrylic acid (L0)**. Kitamura discloses in Table 1, the **emission of dye 1b is around 550nm or yellow-green**. Kakiage discloses use of **LEG4 adsorbed onto a TiO_2 electrode** to improve electron injection and conversion efficiency. Thus, it would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to substitute the LEG4 and 4-diphenylamino phenylcyanoacrylic acid (L0) dyes for cyan and yellow, respectively, along with use of SQ2 as taught by Zheng into the composition and display of Zhang to produce a photochromic liquid for use in a display. Therefore, Zhang, Zheng, Kitamura, and Kakiage teach the claimed "**The display of claim 7 wherein the colloid is a TiO2 colloid and the dyes are: (a) SQ2 (5-carboxy-2-[[3-[(2,3-dihydro-1,1-dimethyl-3-ethyl-1H-benzo[e]indol-2- ylidene)methyl]-2-hydroxy-4-oxo-2-cyclobuten-1-ylidene]methyl]-3,3-dimethyl-1-octyl-3H- indolium),(b) LEG4 (3-{6-{4-[bis(2',4'-dibutyloxybiphenyl-4-yl)amino-]phenyl}-4,4-dihexyl- cyclopenta-[2,1-b:3,4-**

b']dithiophene-2-yl}-2-cyanoacrylic acid) and (c) L0 (4-(diphenylamino) phenylcyanoacrylic acid)".

14. **Claim 6** is rejected under 35 U.S.C. 103 as being unpatentable over Zhang et al in view of Zheng et al, Kitamura et al, and Kakiage et al as applied to **claim 4** above, and further in view of Kang Uk Lee et al., "Mobility of black pigments for electrophoretic display depending on the characteristics of carbon sphere," Dyes and Pigments, Volume 121, 2015, Pages 276-281, ISSN 0143-7208, https://doi.org/10.1016/j.dyepig.2015.06.001.

a. Regarding **claim 6**, Zhang, Zheng, Kitamura, and Kakiage teach the **chromic liquid of claim 4**. Further, both Zheng and Kitamura describe that these adsorbed dyes can be typically implemented into dye-sensitized solar cells. Zheng and Kitamura both disclose **that these set ups include a redox couple (I^-/I_3^-) in the electrolyte**. Additionally, Zhang describes that **the photochromic dye is "directly dissolved in the transparent paint" and thus contains a solvent**. Kang Uk Lee et al disclose **carbon spheres that can serve as a black pigment** (otherwise an "ink" or "dye") for use in electrophoretic display devices. In the introduction, Uk Lee discloses the basic **components of electrophoretic displays (EPD) which are microcapsules which contain colloidal pigments dispersed in a dielectric field**. For monochromic displays, a **colored particle with a dye solution** was initially used. Later, **more than two types of particles are applied for** color realization and enhancement of contrast. In their implementation, Uk Lee **disperses the particles in a polar or nonpolar solvent**. Therefore, it would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to disperse the colloidal solution of dyes into a solvent for use in

an EPD. Thus, Zhang, Zheng, Kitamura, Kakiage, and Uk Lee teach the claimed "**The chromic liquid of claim 4 further comprising solvents and other redox couples.**".

15. **Claim 7** is rejected under 35 U.S.C. 103 as being unpatentable over Zhang et al in view of Zheng et al and Kang Uk Lee et al.

a. Regarding **claim 7**, as described in the rejection of **claim 1**, Zhang discloses a **photochromic ink (liquid)** composed of **three dyes** corresponding to **cyan, fuschine (also referred to as "red purple dye" which "is called magenta" under the "preparation" section), and yellow** whereby **using different absorption spectrum of each dye to control each color channel in a photochromic coating**. By modifying the light source (see explanation under material 2 in document, preparation of photochromic coating, and Fig. 4), the **full gamut of color output can be obtained**. Furthermore, Zhang discloses use **of a projector which transmits the "texture generated by the computer to the canvas" whereby the canvas is the photochromic liquid/ink**. The **user can "use handheld UV projector on the photo-chromic canvas 'picture'", thus a system for projecting a color image onto the photochromic material**. Under "modomodify projector" section, Zhang describes use of an **existing DLP projector because it already comprises three LED light sources of different colors (RGB)** and **use of an optical filter for each LED between collimators (see Fig. 4, 'transparent' containers depicted for Fig. 5**). Additionally, **Fig. 3 shows emission spectra of the projector which are "spiked" due to the color optical filter**. However, Zhang does not disclose use of a colloid imbued with the dyes. Zheng et al **teach imbuing TiO_2 nanowires with individual organic (RGB) dyes** in a **liquid solution (thus a colloid, Fig. 1) contained in transparent containers** which are also capable of achieving the **full**

color gamut (Fig. 2). Kang Uk Lee et al disclose **carbon spheres that can serve as a black pigment** (otherwise an "ink" or "dye") for use in electrophoretic display devices. In the introduction, Uk Lee discloses the basic **components of electrophoretic displays (EPD) which are microcapsules which contain colloidal pigments dispersed in a dielectric field**. For monochromic displays, a **colored particle with a dye solution** was initially used. Later, **more than two types of particles are applied for** color realization and enhancement of contrast. EPDs are commonly known to contain a **transparent electrode layer, a bottom supporting layer, and a transparent oil**. Uk Lee further discloses that the pigments are contained between two electrodes (otherwise a container housing the suspension), whereby light passes through the **transparent electrode**, reflects on the surface of the particles near the electrode, and the reflected light reaches the human eye. It would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to implement the colloidal suspension in a transparent container for use in an EPD. Further, it would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, in a container transparent through all sides if user desires light to fully pass through the solution as opposed to reflecting back. Thus, Zhang, Zheng and Uk Lee satisfy the "**A color display comprising:a transparent container containing a colloid imbued with at least three colored dyes selected on the basis of a color system;a first projector for projecting a color image onto the container for a fixed period of time; and three light sources of different color which direct their light through color related spike optical filters and then into the container for a certain period of time.**".

16. **Claim 8** is rejected under 35 U.S.C. 103 as being unpatentable over Zhang et al in view of Zheng et al and Kang Uk Lee et al as applied to claim 7 above, and further in view of Kitamura et al and Kakiage et al.

a. Regarding **claim 8**, Zhang, Zheng, and Uk Lee teach the display of **claim 7**. As described in the rejection of **claim 4**, Zheng also **discloses use of a TiO_2 colloid and where one dye is SQ2**. Zheng states that any organic dye may be used. Kitamura characterizes phenyl-conjugated oligoene dyes adsorbed on TiO_2 films. In scheme 1, the **structure of 1b** is that of **4-diphenylamino phenylcyanoacrylic acid (L0)**. Kitamura discloses in Table 1, the **emission of dye 1b is around 550nm or yellow-green**. Kakiage discloses use of **LEG4 adsorbed onto a TiO_2 electrode** to improve electron injection and conversion efficiency. Thus, it would have been prima facie obvious to one of ordinary skill in the art, as of the effective filing date, to substitute the LEG4 and 4-diphenylamino phenylcyanoacrylic acid (L0) dyes for cyan and yellow, respectively, along with use of SQ2 as taught by Zheng into the composition and display of Zhang to produce a photochromic liquid for use in a display. Therefore, Zhang, Zheng, Uk Lee, Kitamura, and Kakiage teach the claimed "**The display of claim 7 wherein the colloid is a TiO2 colloid and the dyes are: (a) SQ2 (5-carboxy-2-[[3-[(2,3-dihydro-1,1-dimethyl-3-ethyl-1H-benzo[e]indol-2- ylidene)methyl]-2-hydroxy-4-oxo-2-cyclobuten-1-ylidene]methyl]-3,3-dimethyl-1-octyl-3H- indolium),(b) LEG4 (3-{6-{4-[bis(2',4'-dibutyloxybiphenyl-4-yl)amino-]phenyl}-4,4-dihexyl- cyclopenta-[2,1-b:3,4-b']dithiophene-2-yl}-2-cyanoacrylic acid) and (c) L0 (4-(diphenylamino) phenylcyanoacrylic acid)**".

Conclusion

17. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Eshkalak et al provides a review on electronic inks utilized in electronic displays (relevant to claims 7-8).

18. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Noa W. F. Grooms whose telephone number is (571)272-9981. The examiner can normally be reached M-F 7:30-3:30PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Curtis Mayes can be reached at (571) 272-1234. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC)

at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service

Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/NWFG/
Examiner, Art Unit 1759

/MELVIN C. MAYES/
Supervisory Patent Examiner, Art Unit 1759